```
/*7. Write C++ program to implement
Cohen Southerland line clipping
algorithm.
*/
```

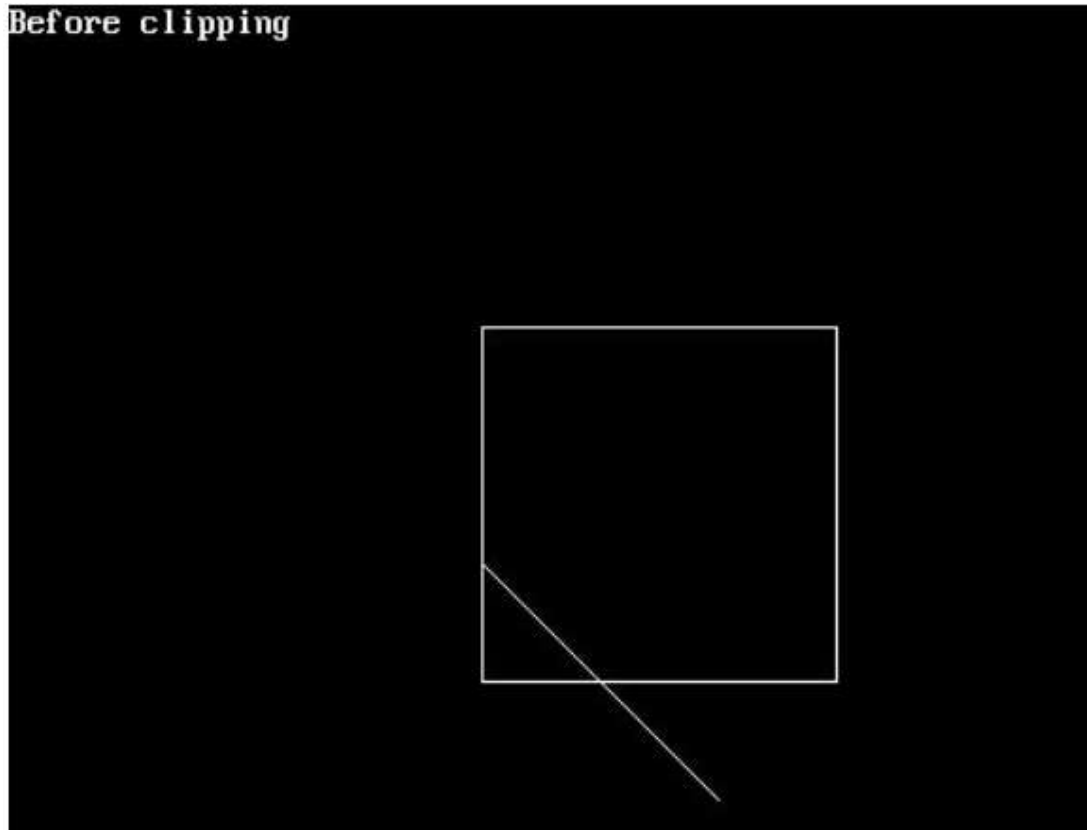
CODE:

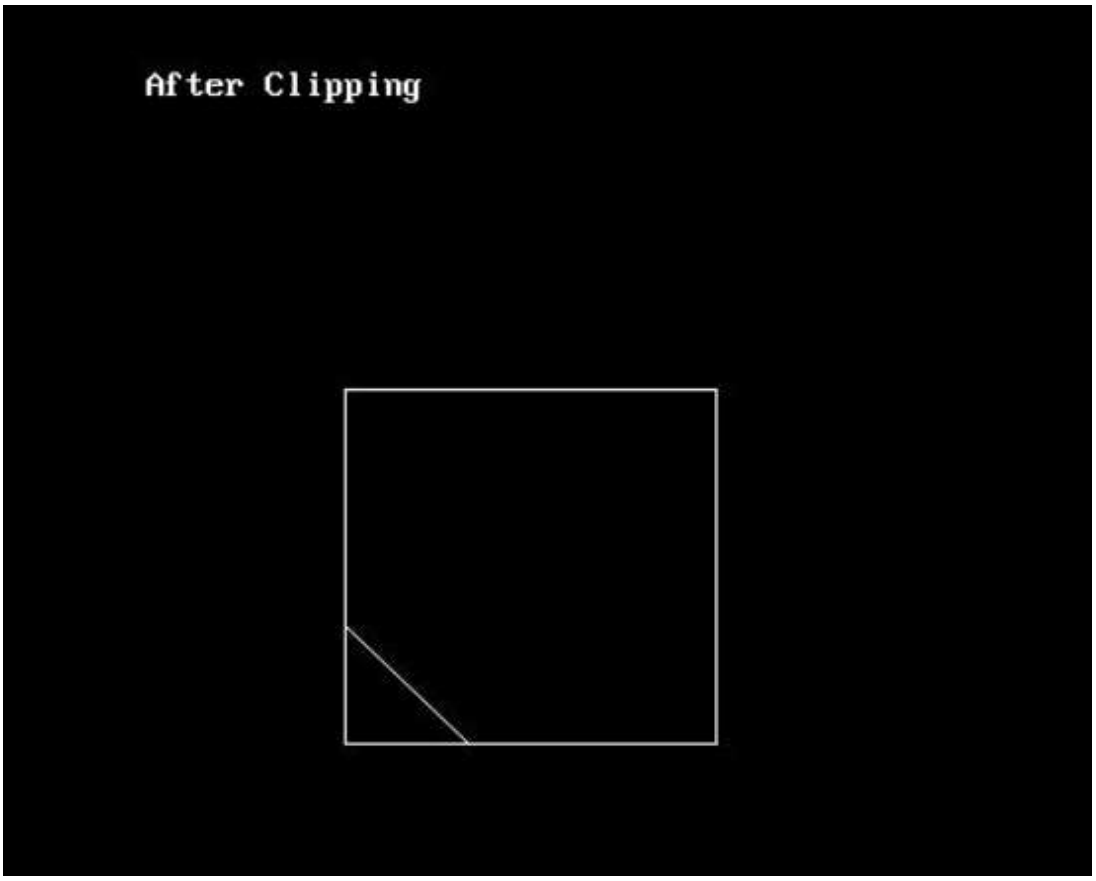
```
#include<iostream.h>
#include<conio.h>
#include<graphics.h>
#include<math.h>
void Window()
{
   line (200,200,350,200);
   line(350,200,350,350);
   line(200,200,200,350);
   line(200,350,350,350);
}
void Code(char c[4],float x,float y)
{         c[0]=(x<200)?'1':'0';
    c[1]=(x>350)?'1':'0';
    c[2]=(y<200)?'1':'0';
    c[3]=(y>350)?'1':'0';
}
void Clipping  (char c[],char d[],float
&x,float &y,float m)
{
   int flag=1,i=0;
   for (i=0;i<4;i++)
   {
      if(c[i]!='0' && d[i]!='0')
      {
         flag=0;
```

```
            break;
        }
        if(flag)
        {
            if(c[0]!='0')
            {
                y=m*(200-x)+y;
                x=200;
            }
            else if(c[1]!='0')
            {
                y=m*(350-x)+y;
                x=350;
            }
            else if(c[2]!='0')
            {
                x=((200-y)/m)+x;
                y=200;
            }
            else if(c[3]!='0')
            {
                x=((350-y)/m)+x;
                y=350;
            }
        }
        if (flag==0)
            cout<<"Line lying outside";
    }
}
void main()
{
int gdriver = DETECT, gmode, errorcode;
float x1,y1,x2,y2;
float m;
```

```
char c[4],d[4];
clrscr();
initgraph(&gdriver, &gmode,
"//Turboc3//bgi");
cout<<"Enter coordinates";
cin>>x1>>y1>>x2>>y2;
cout<<"Before clipping";
Window();
line(x1,y1,x2,y2);
getch();
cleardevice();
m=float((y2-y1)/(x2-x1));
Code(c,x1,y1);
Code(d,x2,y2) ;
Clipping(c,d,x1,y1,m);
Clipping(d,c,x2,y2,m);
cout<<"After Clipping";
Window();
line(x1,y1,x2,y2);
getch();
closegraph();
}
```

OUTPUT:




After Clipping